Chapter 6 • Trigonometry

Solved Exercise 6.5

1. Find x, y and z in the four right triangles.

(i) A = 30°, y
= 4 is adjacent: x
= 4tan30°
= $\frac{4}{\sqrt{3}}$
z = $\frac{4}{\cos 30°}$
= $\frac{8}{\sqrt{3}}$
x=$\frac{4}{\sqrt{3}}$, z=$\frac{8}{\sqrt{3}}$

(ii) A = 45°, y
= √3 is opposite: x
= y
= √3
z = $\frac{y}{\sin 45°}$
= √6.
x=√3, z=√6

(iii) C=60°, z=2 is hypotenuse: x=zcos60°=1; y=zsin60°=√3.
x=1, y=√3

(iv) A = 45°, y
= 4 is adjacent: x
= 4
z = $\frac{4}{\cos 45°}$
= 4√2.
x=4, z=4√2.

2. Find the unknown side and acute angles.

(i) b = √[(√13)²+(√3)²]
= 4 cm. A
= $\tan^{-1}(\frac{\sqrt{3}}{\sqrt{13}})$
= 25.64°, so C
= 64.36°.
b=4 cm, A=25.64°, C=64.36°

(ii) a=c=4 cm, so b=4√2 cm and A=C=45°.
b=4√2 cm, A=C=45°.

3. Each side of a square field is 60 m. Find both diagonals.

Each diagonal=60√2 m≈84.85 m; both diagonals are equal.

4. Solve each triangle with B=90°.

(i) C = 60°, c
= 3√3: A
= 30°, b
= $\frac{c}{\sin 60°}$
= 6, a
= $\frac{c}{\tan 60°}$
= 3.

(ii) C = 45°, a
= 8: A
= 45°, c
= 8, b
= 8√2.

(iii) a = 12, c
= 6: b
= √180
= 6√5
A = $\tan^{-1}2$

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= 63.43°, C
= 26.57°.
(iv) A = 60°, c
= 4: C
= 30°, b
= 8, a
= $4\sqrt{3}$.
(v) A = 30°, c
= 4: C
= 60°, b
= $\frac{8}{\sqrt{3}}$, a
= $\frac{4}{\sqrt{3}}$
(vi) b = 10, a
= 6: c
= 8, A
= $\sin^{-1}(\frac{3}{5})$
= 36.87°, C
= 53.13°.

5. Canal triangle: PQ=13 m, QR=5 m and PR is perpendicular to QR. Find canal width and ∠PQR in radians.

PR = $\sqrt{(13^2-5^2)}$
= 12 m.
∠PQR = $\cos^{-1}(\frac{5}{13})$
= 1.1760 rad.
Width 12 m; angle 1.18 rad approximately.

6. In the adjoining figure, AB=10 cm, AD=17 cm, CD=8 cm and both ABD and BCD are right triangles. Find x=BC.

$BD^2 = AD^2-AB^2$
= 289−100
= 189.
$x^2 = BD^2-CD^2$
= 189−64
= 125.
x=√125=5√5 cm≈11.18 cm (not 55 cm; the radical is essential).

7. A ladder of length 8 m has its foot 2 m from the wall. Find the height reached.

h = $\sqrt{(8^2-2^2)}$
= $\sqrt{60}$
= $2\sqrt{15}$ m.
7.75 m approximately.

8. A rectangle has diagonal (x+9) m and sides (x+7) m and x m. Find x.

$(x+9)^2=(x+7)^2+x^2$
⇒ $x^2-4x-32=0$
⇒ $(x-8)(x+4)=0$.
A length is positive, so x=8.
x=8 m.

9. Calculate x in each right-triangle case.

(i) x = $\sqrt{(20^2-12^2)}$
= $\sqrt{256}$
= 16 cm.
(ii) AB=BC=4 cm and DC=5 cm. In triangle DBC, DB=$\sqrt{(5^2-4^2)}$=3 cm.
Then in triangle ABD, x=AD=$\sqrt{(AB^2+DB^2)}$=$\sqrt{(4^2+3^2)}$=5 cm.

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